



Ostad
Learn Skill Live

MODULE 16 ASSIGNMENT

Submitted To:

Ostad - Learn Skills Live Course
Topic: JavaScript

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1. Variables & Scoping What is the difference between var, let, and const in JavaScript?

Answer: In JavaScript, var, let, and const are used to declare variables, and they differ mainly in their scope and whether they can be changed.

- ❖ **var** : function-scoped and can be redeclared.
- ❖ **let** : block-scoped and can be reassigned.
- ❖ **const** : block-scoped but cannot be reassigned.

Example:

```
var x = 10;  
  
let y = 20;  
  
const z = 30;  
  
y = 25; // Allowed  
  
// z = 35; // Error
```

2. Operators What is the difference between == and ===? Give one example.

Answer: The main difference between == and === are:

==	===
i. Loose equality operator.	i. Strict equality operator.
ii. Checks value only.	ii. Checks value and data type.
iii. May perform type conversion .	iii. Does not perform type conversion.
iv. Example: console.log(5 == "5"); // true	iv. console.log(5 === "5"); // false

3. Loops What is the difference between a for loop and a for-of loop?

Answer: The Difference between a for loop and a for-of loop are:

for Loop	For-of Loop
i. Used when we need control over the index and condition.	i. Used to directly access the values of an iterable.
ii. Usually works with an index (i).	ii. Does not require an index.
iii. Gives more control over the loop.	iii. Simpler and easier to read.
iv. Example: <code>for (let i = 0; i < arr.length; i++)</code>	iv. Example: <code>for (let item of arr)</code>

4. Functions What is an arrow function? Write a simple example.

Answer: An arrow function is a shorter and simpler way to write a function in JavaScript. It uses the `=>` symbol.

Example :

```
const add = (a, b) => a + b;
```

```
console.log(add(5, 3)); // 8
```

5. Advanced Functions What are an IIFE, a recursive function, and a generator function? Write one key characteristic of each.

Answer:

IIFE : A function that is executed immediately after it is created.

Key Characteristic : Runs immediately without being called separately.

Recursive Function : A function that calls itself to solve a problem.

Key Characteristic : Calls itself and needs a base condition to stop.

Generator Function : A special function that can pause and resume its execution.

Key Characteristic : Uses `yield` to return values one at a time.

Examples:

```
// IIFE
```

```
(function () {  
    console.log("Hello");  
})();
```

```
// Recursive Function
```

```
function count(n) {  
    if (n === 0) return;  
    console.log(n);  
    count(n - 1);  
}
```

```
// Generator Function
```

```
function* numbers() {  
    yield 1;  
    yield 2;  
}
```